Suyog Neupane

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Education

Bachelor's in Electrical Engineering (Honors): University of Texas Arlington 08/2023 – Class of 2027
GPA - 3.67/4

Experience

Research Assistant 02/2024 – present
University of Texas at Arlington Research Institute(UTARI)
- Researching soft robotic systems for upper-body motion assistance and rehabilitation.
- Developed embedded software in **C++** for precise finger, wrist, and hand control.
- Designed, modeled, and tested soft pneumatic wrist exoskeletons using **MATLAB/Simulink**.
- Co-authored three **IEEE papers**.

Robotics Club 02/2024 – 01/2025
AutonoBots_UTA
- Built navigation system for **The Boring Company** competition using **ROS2**, NAV2, and Rviz on a Clearpath Jackal.
- Accelerated YOLO model inference on Jetson Nano using **CUDA** and distributed computing.

NASA LSpace' 09/2023 – 01/2024
Research Lead, Lspace
- Led a research team on commercialization of RFID technology for space logistics.
- Created **CAD** designs in Siemens NX for system modeling and prototype planning.

Avionics System Development, Rocket Project 08/2022 – 08/2023
Squad of Chage Makers Nepal
- Designed **Arduino**-based avionics for amateur rocket with IMU, barometric, and SD data logging.
- Implemented parachute release mechanism and data acquisition for flight parameters.

Awards

The Genspiration Prize 2026
National Academy of Inventors
Designed and developed control algorithm for a novel 2-DoF Wrist exoskeleton, winning **$5000**.

Space Teams University – Mars Autonomous Rover Rally Design Challenge 2025
•Designed and implemented AI Mission Manager algorithms in **Python** for Mars rover control in a high-fidelity simulation (Space Teams PRO).
•Got 4th Position among university teams(Four Countries) winning **$2500.**

Publications

A Novel Design of Two DoFs Soft Pneumatic Wrist Exoskeleton for Rehabilitation 2026

Experimental Data-Driven Angular Velocity Control for a Soft Pneumatic Wrist Exoskeleton for Motion Assistance in Rehabilitation 2025

Learning-based Approach to Assist Occupational Workers using Soft Pneumatic Wrist Exoskeleton 2025

Skills

Technical Skills
C\C++, MATLAB/Simulink, Circuit Design ,Arduino IDE, ESP 32, Multisim, Python, SOLIDWORKS, CUDA, NI LabVIEW, 3D Printing, Solder, ROS